

Project requirements

1) Core goals

- Let a user **track what's in the fridge**
- Let a user **track expiry dates** (at least "use soon")
- Let a user **manage a shopping list**
- Make it **fast** to add/remove items

2) MVP functional requirements

A. Authentication & user account

- Users can **create an account, log in, log out**
- Users can **reset password**
- User data is **private per account**

B. Fridge inventory

- User can **add an item** to fridge
 - Fields: name, quantity, unit (optional), category (optional), expiry date (optional), notes (optional)
- User can **edit an item**
- User can **delete an item**
- User can **mark item as "used up"** (1 tap) and then it will be removed
- User can **view fridge list (items in the fridge)**
 - Sort by: expiry date (default), name, category
 - Filter: "expiring soon", category

C. Expiry tracking

- App highlights items in states:

- **Expiring soon** (e.g., within 2 days)
- **Expired** (past date)
- User can set expiry date when adding/editing
- (Optional but helpful) User can set a **default shelf-life** per category/item

D. Shopping list

- User can **add item to shopping list**
 - Fields: name, quantity (optional), notes (optional)
- User can **check off** an item
- User can **remove** item from list
- User can **reorder** items (optional)
- User can move item:
 - **Shopping list** → **Fridge** (1 tap "Bought" adds to fridge)

E. Basic "smart" actions (still simple)

- From a fridge item: **"Add to shopping list"**
- From a shopping item: **"Add to fridge"**
- Prevent duplicates gently:
 - If item name matches existing one, prompt: "Increase quantity or create new?"

3) Non-functional requirements (MVP)

- **Speed:** common actions (add/check/delete) should take < 3 taps
 - **Offline support:** app works without internet; sync later if logged in
 - **Data storage:** persistent local storage (and cloud if account-based)
 - **Security:** password stored securely; HTTPS for all network traffic
 - **Reliability:** no data loss on app restart
 - **Accessibility:** readable font sizes, good contrast, supports screen readers (basic)
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4) Data model requirements (minimum)

FridgeItem

- id
- userId (if accounts)
- name (string)
- quantity (number, optional)
- unit (string, optional)
- expiryDate (date, optional)
- category (string, optional)
- createdAt, updatedAt

ShoppingItem

- id
 - userId
 - name
 - quantity (optional)
 - checked (boolean)
 - createdAt, updatedAt
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5) Screens (minimum)

- Login/Register (if using accounts)
 - **Fridge** screen (list + add button)
 - **Add/Edit Item** screen
 - **Shopping List** screen
 - Settings (minimal: expiry threshold days)
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6) Out of scope for MVP (explicitly)

- Receipt scanning / barcode scanning
- Recipe suggestions

- Multi-user household sharing
- Nutrition tracking
- Price tracking or store integration
- AI recognition of food from camera